

Air Quality Health Index (AQHI) — Edmonton

2.0

Current AQHI at Edmonton McCauley (0.84 km from venue)
Forecast Sat 3:00 PM at Edmonton McCauley (0.84 km from venue): **2.5**

Health Risk	AQHI	At-Risk Population	General Population
Low Risk	2.5	Enjoy your usual outdoor activities.	Ideal air quality for outdoor activities.

No active Environment Canada weather alerts for the venue.

Environmental Intelligence Summary

- At Edmonton Symphony Orchestra Outdoor Concert, temperature is 19.1°C and feels near 18.7°C. Winds are 4 km/h from SE, gusting near 15 km/h.
- The nearest current AQHI is 2 at Edmonton McCauley, 0.8 km from the venue.
- The official Government of Alberta AQHI for Edmonton is 2 (forecast tonight: 3, tomorrow: 3).
- A gridded AQHI estimate blending official and community sensors (confidence: High, nearest contributing point 1.6 km away) puts the area near 1.
- The nearest air monitoring station (Edmonton McCauley, 0.8 km) reports fine particulate matter (PM2.5) at 8.1 µg/m³.
- A nearby community sensor ('RE/MAX Field', 1.2 km) reads 6.8 µg/m³ PM2.5.
- Back-trajectory modeling (HRDPS wind fields, 6 h lookback) indicates the air currently at the venue traveled from the vicinity of a point 171.2 km to the W, i.e. arriving from the W.
- The nearest active fire detection (NASA FIRMS – VIIRS, last 2026-08-28) is 7.0 km E of the venue. The modeled back-trajectory passes within 13.4 km of this fire, making smoke transport to the venue plausible based on wind-field modeling.
- The AQHI forecast for Sat 3:00 PM is 2, based on Edmonton McCauley, 0.8 km from the venue.
- Precipitation probability reaches approximately 59%.
- Overall operational environmental risk is MODERATE.

Recommendations

- Environment Canada AQHI guidance — general population: Ideal air quality for outdoor activities.
- Environment Canada AQHI guidance — at-risk populations: Enjoy your usual outdoor activities.

Hazard Snapshot — Overall Risk: Moderate Risk

AIR QUALITY Low Risk 2.5 AQHI	HEAT Low Risk 21.1 °C apparent/humidex	WIND Low Risk 40.3 km/h peak gust	PRECIPITATION Moderate Risk 2.9 mm/h maximum	THUNDERSTORM Low Risk — first forecast signal
AQHI RATE OF CHANGE Low Risk 0.0 AQHI change/hour			WIND SHEAR Low Risk 1.4 ° direction diff (10m vs 120m)	

Local Monitoring

Nearest station 2.0 Edmonton McCauley · 0.84 km	Blended estimate 1.0 confidence High	PM2.5 (station) 8.1 µg/m³ Edmonton McCauley · 0.85 km	PM2.5 (community) 6.8 µg/m³ RE/MAX Field · 1.21 km
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